

Sri Datta Ganesh Bandreddi

Data & Platform Engineer | petabyte-scale pipelines, distributed systems, and cost-efficient infrastructure

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00

Nagpur, India

Jul 2022

SKILLS

Languages: Python, Go, Java (Spring Boot), C++, SQL, Bash /Shell Scripting

Data & ETL: Ray (Data), Spark/SparkML, Iceberg, Airflow, NVTabular, Rapids (cuDF, cuML), Pandas, NumPy

Databases: PostgreSQL, Neo4j, JanusGraph, Cassandra, MongoDB, Redis, Kafka, Elasticsearch, FAISS

Systems & MLOps: Docker, Kubernetes, Terraform, Helm, Jenkins/GitHub Actions, AWS (S3, RDS, ECS, EventBridge), GCP

Observability: Prometheus, Grafana, OpenTelemetry, Nsight Systems, MLflow

ML & DL: PyTorch, TensorFlow, Hugging Face, LangGraph, LoRA/QLoRA

PROFESSIONAL EXPERIENCE

MOTIONAL

Data Platform Intern (Infrastructure)

Pittsburgh, PA

May 2026 - Aug 2026

- Cut drivelog thaw costs by about \$100K/month, by architecting a metadata-extraction platform (PoC to production on a medallion architecture) that answers characterization queries on petabyte-scale logs without thawing from cold storage
- Selected and shipped the production data stack, by benchmarking Ray Data against Spark on Kubernetes and building ingestion pipelines on AWS S3, RDS, and EventBridge
- Accelerated region-level analytics and enabled precise, timeboxed thawing of long-tail events, by enriching drivelogs with weather and map layers and building H3 geospatial rollups

PHENOM

Product Development Engineer (Enterprise Talent Graph)

Hyderabad, India

Dec 2023 - Jul 2025

- Scaled a production ontology-driven knowledge graph to 1M+ entities with continuous ingestion, by designing validated aggregation pipelines on Neo4j graph schemas and Iceberg versioned storage
- Ensured high availability for workloads serving 100+ enterprise clients, by managing distributed Kafka, PostgreSQL, and Redis clusters and building CI/CD pipelines with Jenkins and Helm charts on Kubernetes
- Benchmarked 8 databases for knowledge-graph workloads and cut AI-driven query response time by 40%, by evaluating scalability and storage efficiency and tuning PostgreSQL horizontal sharding

BYJU'S

Software Engineer (Core Platform Team)

Hyderabad, India

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% for 1M+ users, by deploying scheduled AWS ECS auto-scaling and Terraform IaC tuned to traffic patterns
- Halved reliance on external identity providers, by leading development of an in-house auth platform (SAML, SSO, RBAC) in Go and Java (Spring Boot)

PHENOM

Machine Learning Intern

Hyderabad, India

May 2021 - Jun 2022

- Cut recommender preprocessing latency, by migrating CPU data pipelines to NVTabular on GPU and validating scalability across 2 production databases
- Streamlined database migration and testing, by building reusable database package clients that standardized DB query interfaces across services

PROJECT EXPERIENCE

FLARE: Efficient Attention (Carnegie Mellon University) (link)

Aug 2025 - Present

- Co-authored a linear-attention operator (3rd of 7 authors, submitted to TMLR) later adopted by an NVIDIA and General Motors crash-dynamics effort for roughly 2x lower memory
- Demonstrated over 200x speedups on Long Range Arena and extended FLARE to image classification and diffusion, by profiling multi-GPU NCCL training across FP8/BF16 with Nsight Systems

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B-parameter LLM to stable convergence with about 40% less memory on four H100 GPUs, by configuring tensor and pipeline parallelism, gradient checkpointing, and LoRA/QLoRA adapters
- Raised training throughput to 2.1x over naive data parallelism, by tuning NCCL, gradient bucketing, and optimizer-state sharding

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Reached 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification (beating ResNet-50, EfficientNet-B3, and ViT baselines), by training a 210M-parameter Swin Transformer
- Cut inference latency by 3x versus PyTorch eager, by converting to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and serving on Triton with adaptive batching

AWARDS & HONORS

Phenomenal Award for Outstanding Performance, Phenom (2024, Q1 & Q2) • 2nd Place, AppLovin Track, TartanHacks 2026 • Provisional Patent, Motional (2026)